

Lab 4 Supplemental – Manufacturing Facility Design and Simulation

◆ 1. ORDER STATISTICS

✓ Preorders waiting time distribution (min.)

- Time orders are spent **waiting before processing begins**.
- Affected by:
 - Availability of goods (from goods statistics)
 - Resource availability (e.g., pickers, conveyors)

✓ Order picking time distribution (min.)

- Time it takes to **pick items** in an order once processing begins.
- Depends on:
 - Number of goods per order
 - Layout/efficiency of the picking process
 - Resource availability

✓ Full order processing time distribution (min.)

- Total time from when the order enters the system to when it's complete.
- = Preorder waiting time + Picking time + Other processing (e.g., packing)

✓ Order states per minute

- Shows the **real-time status** of orders (e.g., waiting, being picked, done)
- Directly reflects the system's throughput and bottlenecks.
- Related to:
 - Conveyor and picking efficiency
 - Availability of goods and resources

✓ Amount of goods per order

- Average or distribution of **how many items/orders** contain.
- Influences:

- Picking time
 - Storage/retrieval complexity
 - Goods availability
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◆ 2. GOODS STATISTICS

✓ Unavailable goods per type

- Goods that are **needed for orders** but are not currently in inventory.
- Impacts:
 - Preorder waiting time (orders wait for goods)
 - Throughput of the system

✓ Stored unreserved goods per type

- Goods in storage **not yet reserved** for any order.
- Indicates:
 - Inventory levels
 - Potential buffer for upcoming orders

✓ Goods in processed orders per type

- Goods are currently being picked or processed as part of active orders.
- Correlates with:
 - Order states: “in process”
 - Picking system load

✓ Stored goods states per minute (reserved, unreserved)

- Time-based snapshot of how many goods are **reserved** (for orders) vs **free**.
- Shows how inventory is utilized over time.
- Tied to:
 - Incoming orders and reservation logic

- Order picking speed

✓ Goods storing time distribution (min.)

- Time goods spend in storage before being used or shipped.
- Can be interpreted as:
 - Inventory turnover
 - Efficiency in order matching and picking

RELATIONSHIPS BETWEEN ORDER AND GOODS STATISTICS

Here's a **visual-like mapping** of the relationships:

[Goods Available (unreserved)] → [Orders Waiting] ↔ [Preorder Waiting Time]

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[Goods Reserved per Order] + [Pickers/Resources]

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[Order Picking Time] → [Goods in Processed Orders]

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[Full Order Processing Time]

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[Stored Goods Reduced] → [Goods Storing Time]

Summary of Key Relationships:

Order Stat	Connected Goods Stat	Relationship
Preorder waiting time	Unavailable goods, Unreserved goods	Orders wait if goods aren't available
Order picking time	Goods per order, Goods in processed orders	More goods = longer picking time
Full order processing time	Combines above	Total impact of goods & process

Order Stat	Connected Goods Stat	Relationship
Order states per minute	Goods in processed orders, Reserved goods	Reflects current load and flow
Amount of goods per order	Impacts on all goods stats	Larger orders tie up more goods
Stored goods states per minute	Tracks how goods are reserved or unreserved	Drives what's available for new orders
Goods storing time	Indicates delays in turnover	Longer storage = slower inventory cycle

Rubric for ENGR-3027 Lab 4 Report

- 1) Manufacturing facility drawing – 4 marks
- 2) AnyLogic part -1- observation and analysis – 2 marks
- 3) AnyLogic part -2- observation and analysis – 2 marks
- 4) AnyLogic part -3- observation and analysis – 2 marks